Air Quality Prediction Using Ensemble Machine Learning and Environmental Data Analysis

A Comparative Study of Statistical and Machine Learning Models for $PM_{2.5}$ Prediction Using the Beijing Multi-Site Air Quality Dataset

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Abstract

Air pollution remains one of the most significant environmental and public health challenges worldwide, with fine particulate matter ($PM_{2.5}$) contributing to substantial respiratory and cardiovascular health risks. Accurate prediction of $PM_{2.5}$ concentrations is therefore important for air quality management, environmental monitoring, and public health decision-making. This report presents a reproducible machine learning framework for predicting hourly $PM_{2.5}$ concentrations using the Beijing Multi-Site Air Quality Dataset, which contains more than 420,000 hourly observations collected across twelve monitoring stations from 2013 to 2017.

The workflow included data integration, preprocessing, feature engineering, exploratory data analysis, supervised machine learning, model comparison, cross-validation, and interactive dashboard deployment. Seven predictive models were evaluated: Linear Regression, Ridge Regression, Lasso Regression, Random Forest Regression, Extra Trees Regression, Gradient Boosting Regression, and Histogram Gradient Boosting Regression. Model performance was assessed using the coefficient of determination (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and five-fold cross-validation.

Among all evaluated algorithms, Extra Trees Regression achieved the strongest predictive performance, with a testing R^2 of approximately 0.946, an MAE of 11.06 $\mu g/m^3$, and an RMSE of 18.66 $\mu g/m^3$. Five-fold cross-validation produced a mean R^2 of 0.9461, a mean MAE of 11.09 $\mu g/m^3$, and a mean RMSE of 18.59 $\mu g/m^3$, indicating strong generalization across independent data partitions. Feature importance analysis showed that PM_{10} was the dominant predictor, followed by carbon monoxide and meteorological variables. The project demonstrates how ensemble learning, statistical evaluation, and reproducible scientific computing can be combined to model large-scale urban air quality data.

Keywords: air quality; $PM_{2.5}$; machine learning; ensemble learning; Extra Trees Regression; Random Forest; environmental data science; Beijing Multi-Site Air Quality Dataset; Python; Streamlit

1 Introduction

Air pollution is one of the leading environmental challenges affecting human health, urban sustainability, and economic development. Among the numerous atmospheric pollutants, fine particulate

matter with an aerodynamic diameter of less than 2.5 micrometers, known as $PM_{2.5}$, is considered especially hazardous because of its ability to penetrate deeply into the respiratory system. Long-term exposure to elevated $PM_{2.5}$ concentrations has been associated with increased risks of cardiovascular disease, respiratory illness, reduced lung function, and premature mortality. Consequently, accurate prediction of $PM_{2.5}$ concentrations has become an important objective within atmospheric science, environmental engineering, and public health research.

Urban air quality is influenced by a complex interaction of anthropogenic emissions, atmospheric chemistry, and meteorological conditions. Pollutant concentrations vary spatially and temporally as a result of changing weather patterns, industrial activity, transportation networks, seasonal variability, and local geographic conditions. These nonlinear interactions make accurate prediction challenging using traditional statistical approaches alone.

Recent advances in machine learning provide new opportunities for modeling complex environmental systems. Ensemble learning algorithms, including Random Forests and Extra Trees Regression, can capture nonlinear relationships without requiring explicit assumptions about functional form. Compared with classical linear regression models, ensemble tree-based methods are able to model complex feature interactions while remaining relatively robust to multicollinearity and noisy observations.

The objective of this project was to develop a reproducible machine learning framework for predicting hourly $PM_{2.5}$ concentrations using the Beijing Multi-Site Air Quality Dataset. The study implements a complete environmental data science workflow consisting of data preprocessing, feature engineering, exploratory data analysis, supervised machine learning, model comparison, cross-validation, and dashboard deployment. The project is intended as a technical portfolio report rather than a mechanistic atmospheric chemistry model.

2 Research Question and Objectives

The central research question was:

> Can pollutant concentrations, meteorological variables, and engineered temporal features be used to accurately predict hourly $PM_{2.5}$ concentrations across multiple air quality monitoring stations?

The specific objectives were:

1. Build a reproducible preprocessing pipeline for the Beijing Multi-Site Air Quality Dataset.
2. Merge twelve station-level datasets into a unified modeling dataset.
3. Engineer temporal features, including season, weekend indicators, and cyclical hour encodings.
4. Conduct exploratory analysis of $PM_{2.5}$ patterns across time, season, station, and predictor variables.
5. Compare seven regression algorithms using identical training and testing datasets.
6. Evaluate model performance using R^2, MAE, RMSE, and five-fold cross-validation.

7. Identify influential predictors through feature importance analysis.

8. Develop an interactive dashboard for model exploration and $PM_{2.5}$ prediction.

3 Dataset and Preprocessing

3.1 Dataset

The analysis used the Beijing Multi-Site Air Quality Dataset, an openly available monitoring dataset distributed through the UCI Machine Learning Repository. The dataset contains hourly air quality observations collected from March 1, 2013 through February 28, 2017 at twelve monitoring stations across the Beijing metropolitan region. Each station records concentrations of major atmospheric pollutants together with meteorological variables.

The original dataset is distributed as twelve separate comma-separated value files, one for each monitoring station. For this project, these files were combined into a single dataset while preserving station identity. After preprocessing, the final analytical dataset contained 420,768 hourly observations.

3.2 Variables

The response variable was hourly $PM_{2.5}$ concentration measured in micrograms per cubic meter. The predictor variables included pollutant measurements, meteorological conditions, and engineered temporal variables. Table 1 summarizes the final modeling variables.

Table 1: Variables used in the $PM_{2.5}$ prediction models.

Variable	Description
PM_{10}	Coarse particulate matter concentration
SO_2	Sulfur dioxide concentration
NO_2	Nitrogen dioxide concentration
CO	Carbon monoxide concentration
O_3	Ozone concentration
TEMP	Air temperature
PRES	Atmospheric pressure
DEWP	Dew point temperature
RAIN	Hourly precipitation
WSPM	Wind speed
Season	Derived seasonal category
Weekend	Binary weekend indicator
Hour sin	Cyclical sine encoding of hour
Hour cos	Cyclical cosine encoding of hour

3.3 Data Cleaning

The preprocessing workflow imported all station-level files, merged them into one dataframe, converted temporal variables into a standardized datetime format, removed unnecessary indexing variables, and verified numerical consistency. Missing numerical observations were imputed using median values, while missing categorical wind-direction values were imputed using the modal category. This produced a complete numerical matrix suitable for regression modeling.

3.4 Feature Engineering

Several additional variables were constructed to improve the representation of temporal patterns. Calendar month was used to define a seasonal variable, and a weekend indicator was generated from the datetime object. Because hour of day is circular, cyclical encoding was applied using sine and cosine transformations:

$$\text{Hour}_{\sin} = \sin\left(\frac{2\pi H}{24}\right), \tag{1}$$

$$\text{Hour}_{\cos} = \cos\left(\frac{2\pi H}{24}\right), \tag{2}$$

where H represents the hour of the day. These transformations preserve the circular structure of time, where 23:00 and 00:00 are adjacent rather than far apart numerically.

4 Methodology

4.1 Analytical Workflow

The analytical workflow consisted of data integration, feature engineering, exploratory visualization, model development, model comparison, cross-validation, and dashboard deployment. The same cleaned dataset and predictor set were used across all models to ensure fair comparison.

4.2 Train-Test Split

To evaluate predictive performance on unseen observations, the dataset was randomly divided into independent training and testing subsets. Eighty percent of the observations were assigned to the training set, while the remaining twenty percent were reserved for evaluation. A fixed random seed was used to ensure reproducibility.

4.3 Machine Learning Models

Seven supervised regression algorithms were implemented and compared. Linear Regression served as the baseline statistical model:

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j + \epsilon. \tag{3}$$

Ridge Regression extends Linear Regression using L2 regularization:

$$\min \left(RSS + \lambda \sum_{j=1}^{p} \beta_j^2 \right), \tag{4}$$

while Lasso Regression uses L1 regularization:

$$\min \left(RSS + \lambda \sum_{j=1}^{p} |\beta_j| \right). \tag{5}$$

Random Forest Regression constructs an ensemble of decision trees trained on bootstrap samples. Predictions are obtained by averaging across individual trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(x), \tag{6}$$

where B is the number of trees and $T_b(x)$ is the prediction from tree b.

Extra Trees Regression is similar to Random Forest Regression but introduces additional randomness during tree construction by choosing split thresholds randomly from candidate values. Gradient Boosting Regression fits trees sequentially so that each new tree reduces the residual error from the existing ensemble:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x), \tag{7}$$

where $h_m(x)$ is a newly fitted regression tree and γ_m is the learning rate. Histogram Gradient Boosting accelerates traditional gradient boosting by discretizing continuous predictor variables into histogram bins.

4.4 Evaluation Metrics

Model performance was evaluated using the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE):

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{8}$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|, \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{10}$$

The R^2 statistic measures the proportion of observed variation explained by the model. MAE gives the average absolute error in $PM_{2.5}$ units, while RMSE penalizes larger prediction errors more strongly.

4.5 Five-Fold Cross-Validation

Five-fold cross-validation was used to assess robustness of the best-performing model. The dataset was divided into five folds. During each iteration, four folds were used for training and the remaining fold was used for validation. This process was repeated five times so that every observation appeared once in a validation set.

5 Results

5.1 Exploratory Data Analysis

Exploratory analysis showed that $PM_{2.5}$ concentrations varied substantially across time, season, and monitoring station. The time series of daily average $PM_{2.5}$ concentrations revealed repeated high-pollution episodes, especially during winter months and early-year periods.

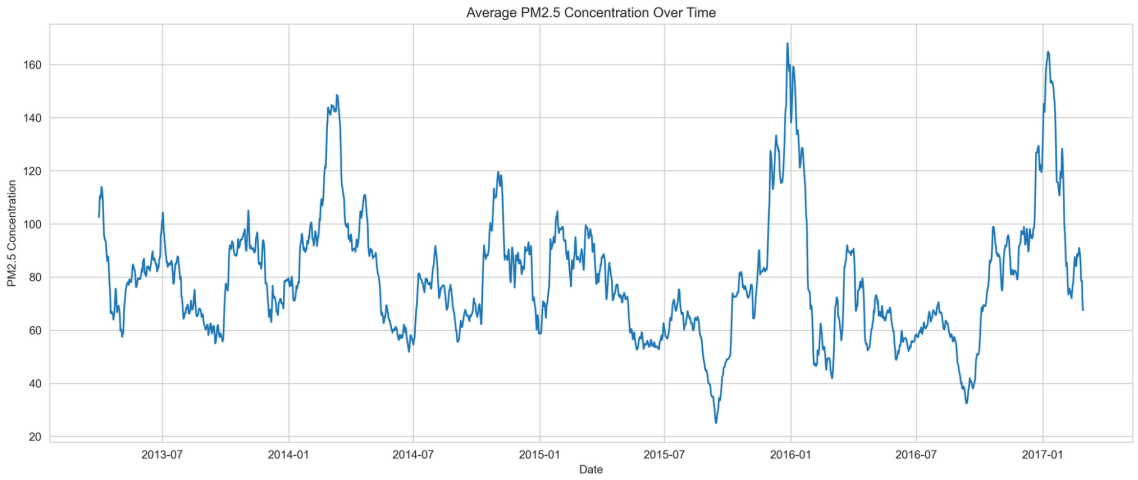


Figure 1: Average $PM_{2.5}$ concentration over time across monitoring stations. The series shows substantial short-term variability and repeated high-pollution episodes.

The distribution of $PM_{2.5}$ concentrations was strongly right-skewed, with many observations at relatively low values and a long tail of severe pollution events. This pattern indicates that the prediction task involves an imbalanced continuous response, where extreme pollution episodes are less common but environmentally important.

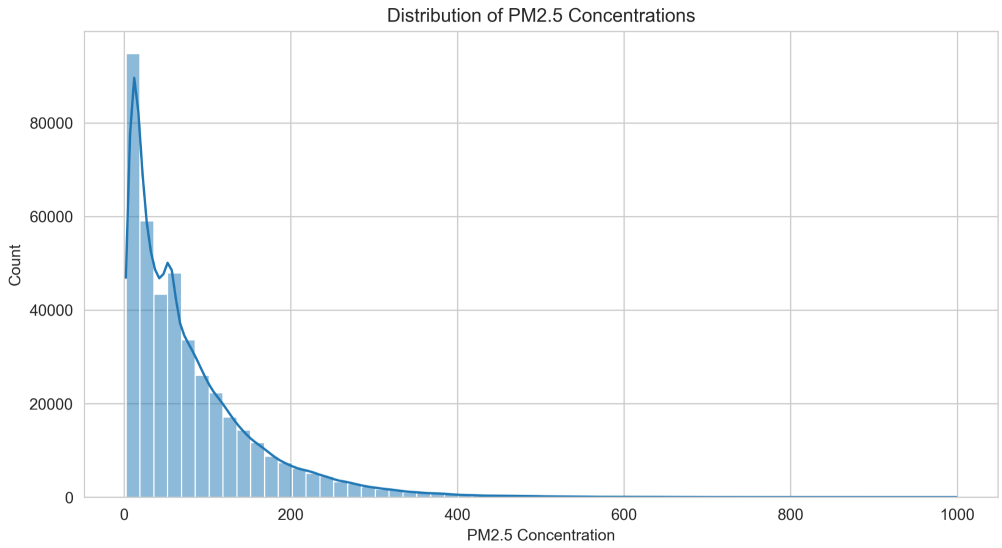


Figure 2: Distribution of hourly $PM_{2.5}$ concentrations. The response variable is strongly right-skewed, with a small number of extreme pollution events.

5.2 Correlation Structure

The correlation matrix showed strong positive associations between $PM_{2.5}$ and PM_{10}, CO, and NO_2. PM_{10} had the strongest correlation with $PM_{2.5}$, consistent with the expectation that particulate matter fractions often share emission sources and atmospheric conditions. Wind speed was negatively associated with $PM_{2.5}$, suggesting that stronger winds may contribute to pollutant dispersion.

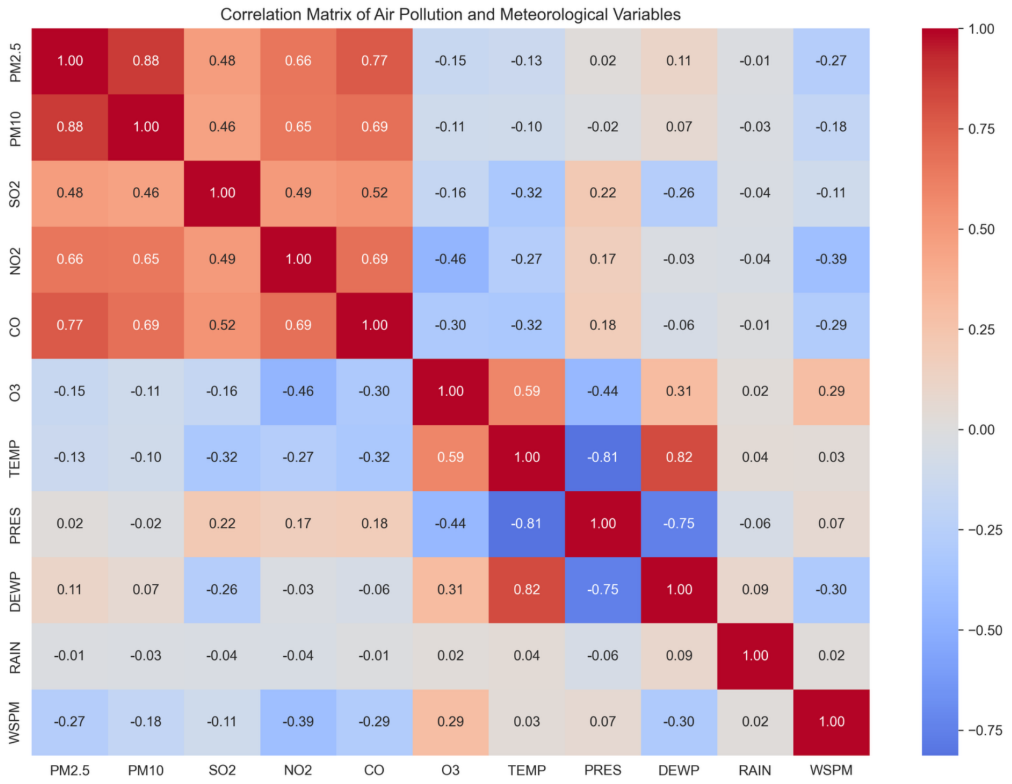


Figure 3: Correlation matrix for air pollution and meteorological variables. $PM_{2.5}$ is strongly correlated with PM_{10}, CO, and NO_2, while wind speed shows a negative association.

5.3 Seasonal and Spatial Patterns

Seasonal distributions showed that winter had the broadest spread and highest extreme $PM_{2.5}$ concentrations. This pattern is consistent with seasonal differences in heating demand, atmospheric stability, and pollutant dispersion. Station-level averages also varied across the monitoring network, indicating spatial heterogeneity in urban air quality.

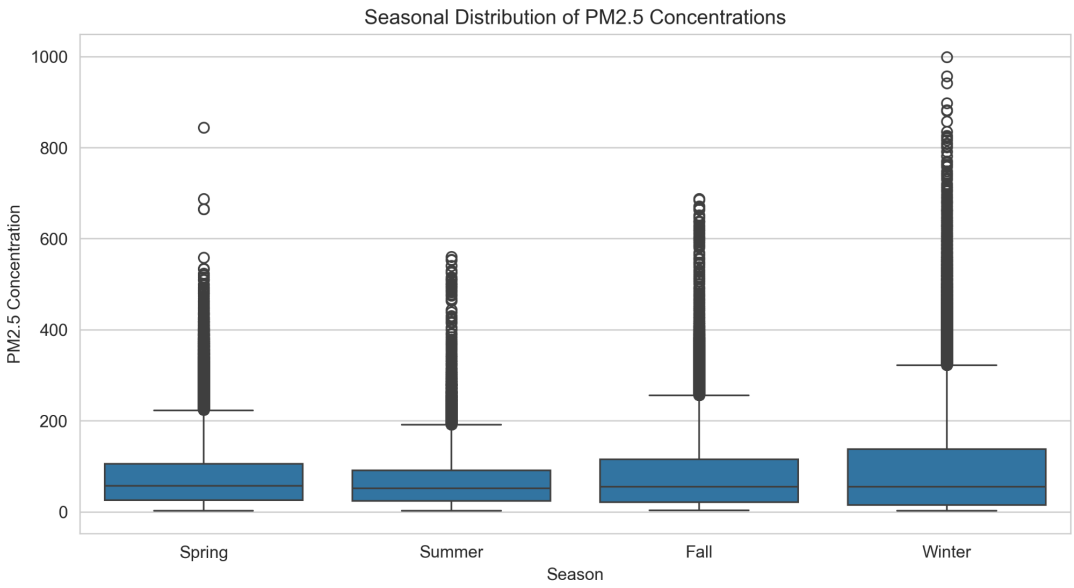


Figure 4: Seasonal distribution of $PM_{2.5}$ concentrations. Winter shows the highest upper tail, indicating more frequent severe pollution episodes.

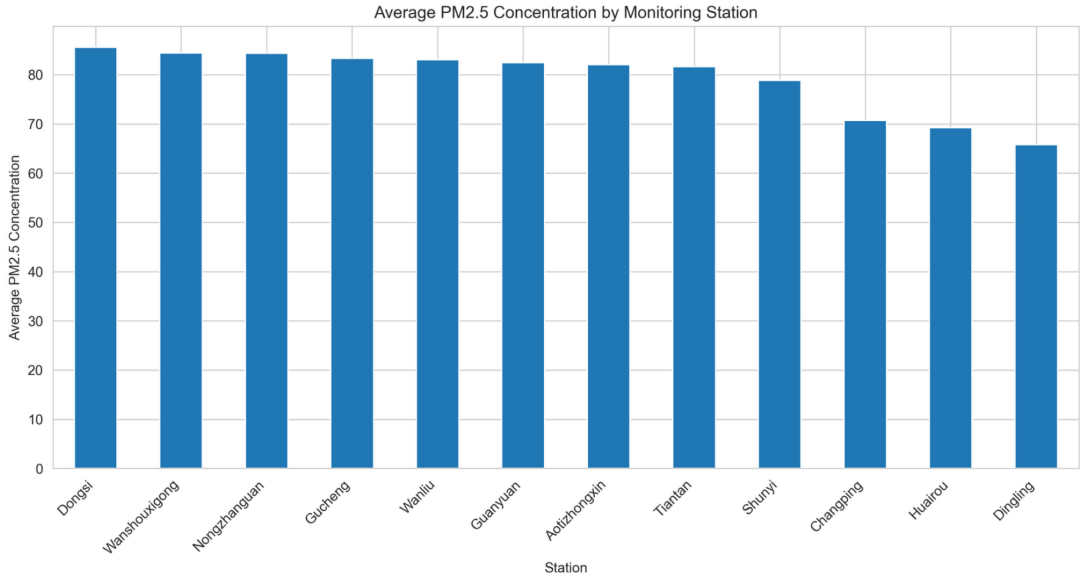


Figure 5: Average $PM_{2.5}$ concentration by monitoring station. Mean concentrations varied across sites, reflecting spatial differences within the monitoring network.

5.4 Model Comparison

The seven regression models showed clear differences in predictive performance. Extra Trees Regression achieved the highest testing performance, followed closely by Random Forest Regression and Histogram Gradient Boosting. The three linear models produced substantially lower predictive performance.

Table 2: Model performance on the held-out testing dataset.

Model	R^2	MAE	RMSE
Extra Trees Regression	**0.9456**	**11.06**	**18.66**
Random Forest Regression	0.9397	11.47	19.65
Histogram Gradient Boosting	0.9261	13.50	21.75
Gradient Boosting Regression	0.9022	15.69	25.02
Linear Regression	0.8418	20.51	31.83
Ridge Regression	0.8418	20.51	31.83
Lasso Regression	0.8418	20.51	31.83

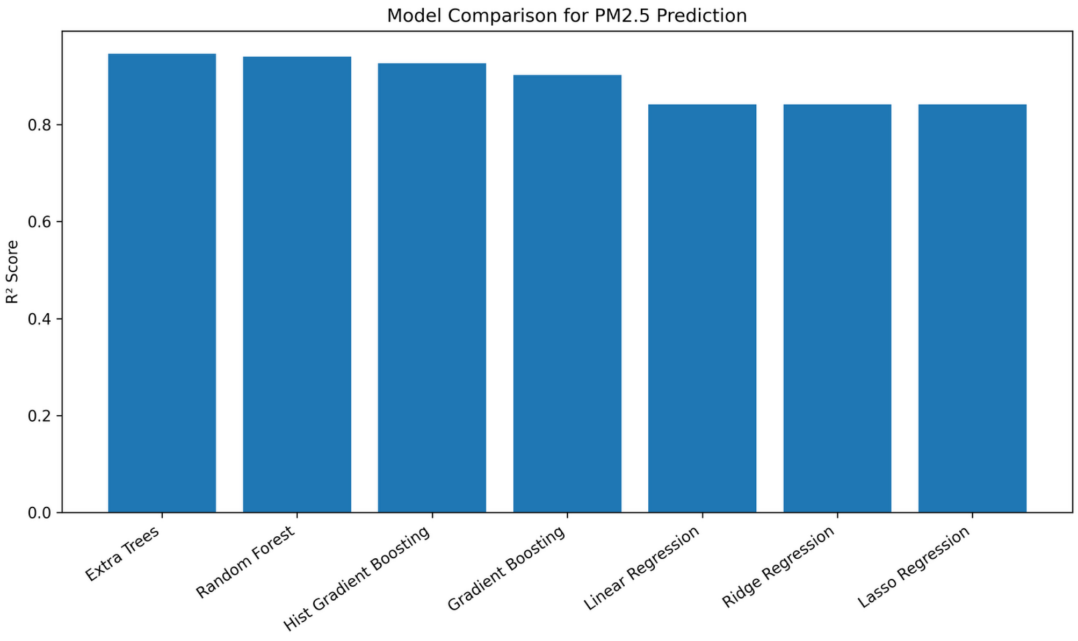


Figure 6: Model comparison using R^2 scores. Extra Trees Regression achieved the highest predictive performance, followed closely by Random Forest Regression.

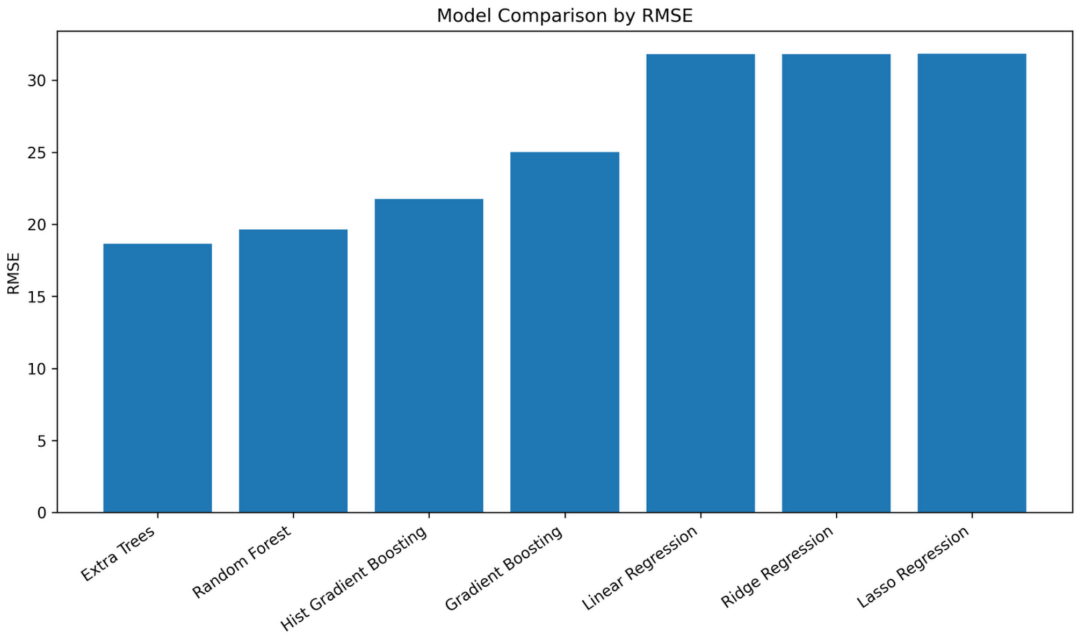


Figure 7: Model comparison using RMSE. Extra Trees Regression produced the lowest prediction error among the evaluated algorithms.

5.5 Cross-Validation

Five-fold cross-validation for the Extra Trees model produced highly consistent validation results. The mean validation R^2 was 0.9461 with a standard deviation of 0.0008. Mean MAE was 11.09 μg/m^3 and mean RMSE was 18.59 μg/m^3.

Table 3: Five-fold cross-validation results for Extra Trees Regression.

Fold	Train R^2	Test R^2	Test MAE	Test RMSE
1	1.0000	0.9457	11.06	18.64
2	1.0000	0.9469	11.06	18.47
3	1.0000	0.9451	11.11	18.74
4	1.0000	0.9469	11.08	18.46
5	1.0000	0.9456	11.12	18.65
Mean	1.0000	**0.9461**	**11.09**	**18.59**

The consistency across folds indicates that the model generalized well across independent subsets of the data. The perfect training R^2 values are expected for high-capacity tree ensembles and should be interpreted alongside the validation performance, which remained strong and stable.

5.6 Prediction Performance and Residuals

The observed-versus-predicted plot showed strong agreement between measured and predicted $PM_{2.5}$ concentrations. Most points clustered near the one-to-one line, although the model displayed more dispersion for extreme pollution values.

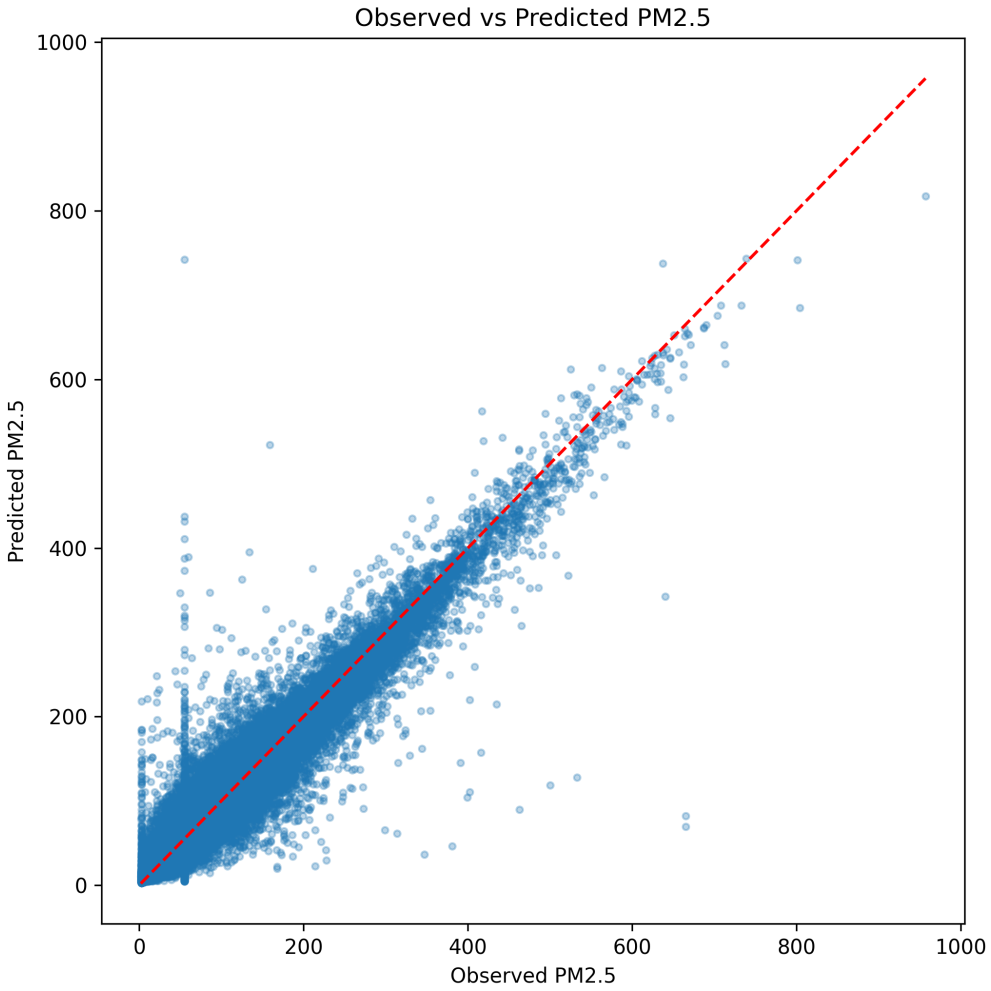


Figure 8: Observed versus predicted $PM_{2.5}$ concentrations for the best-performing model. The dashed line represents perfect prediction.

Residual diagnostics showed that most errors were centered near zero, with a small number of large residuals associated with extreme observations. The residual plot also suggests that prediction uncertainty increases for higher predicted pollution levels, which is common in environmental datasets with rare high-severity events.

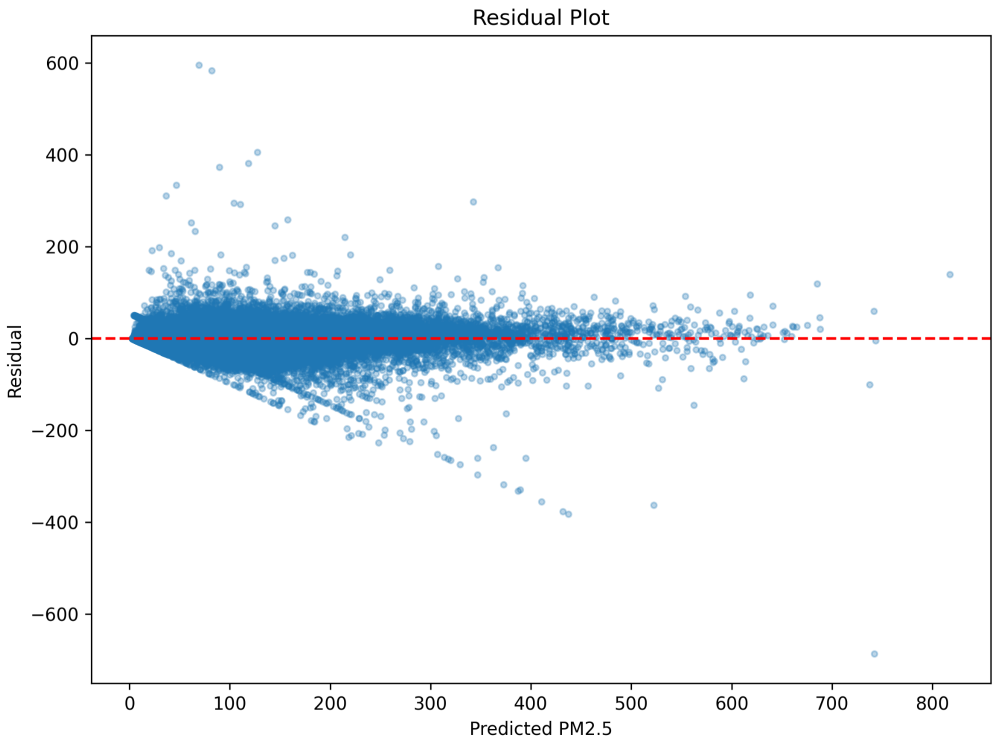


Figure 9: Residual plot for $PM_{2.5}$ predictions. Residuals are centered near zero, but larger errors appear during high-pollution conditions.

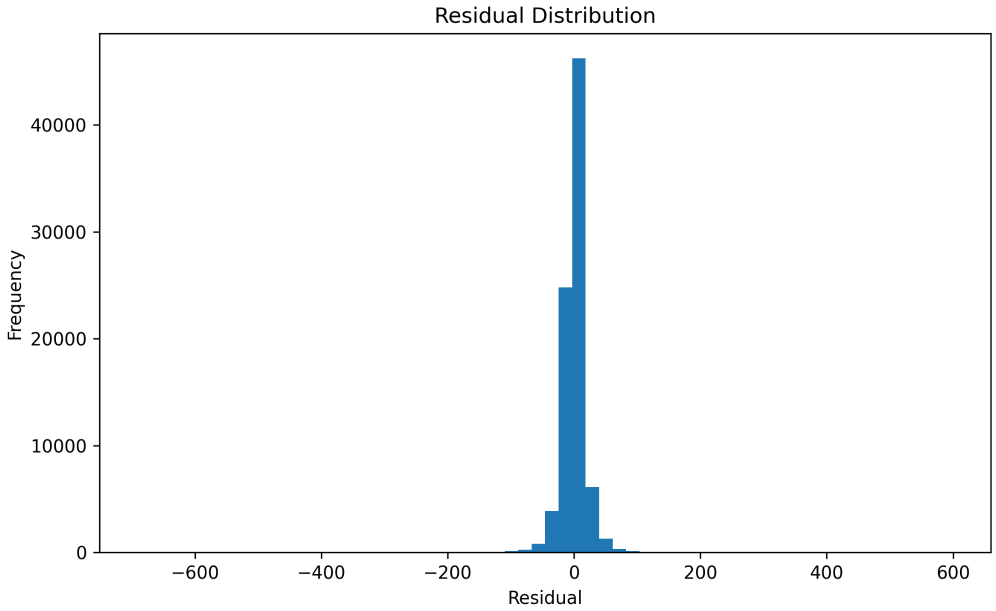


Figure 10: Residual distribution for $PM_{2.5}$ predictions. Most residuals are concentrated near zero, with long tails produced by a small number of difficult high-error cases.

5.7 Feature Importance

Feature importance analysis indicated that PM_{10} was the dominant predictor of $PM_{2.5}$ concentrations. CO, dew point, temperature, SO_2, NO_2, pressure, and ozone contributed smaller but meaningful predictive information. The strong importance of PM_{10} is scientifically reasonable because PM_{10} and $PM_{2.5}$ are related particulate matter fractions and often share sources and atmospheric conditions.

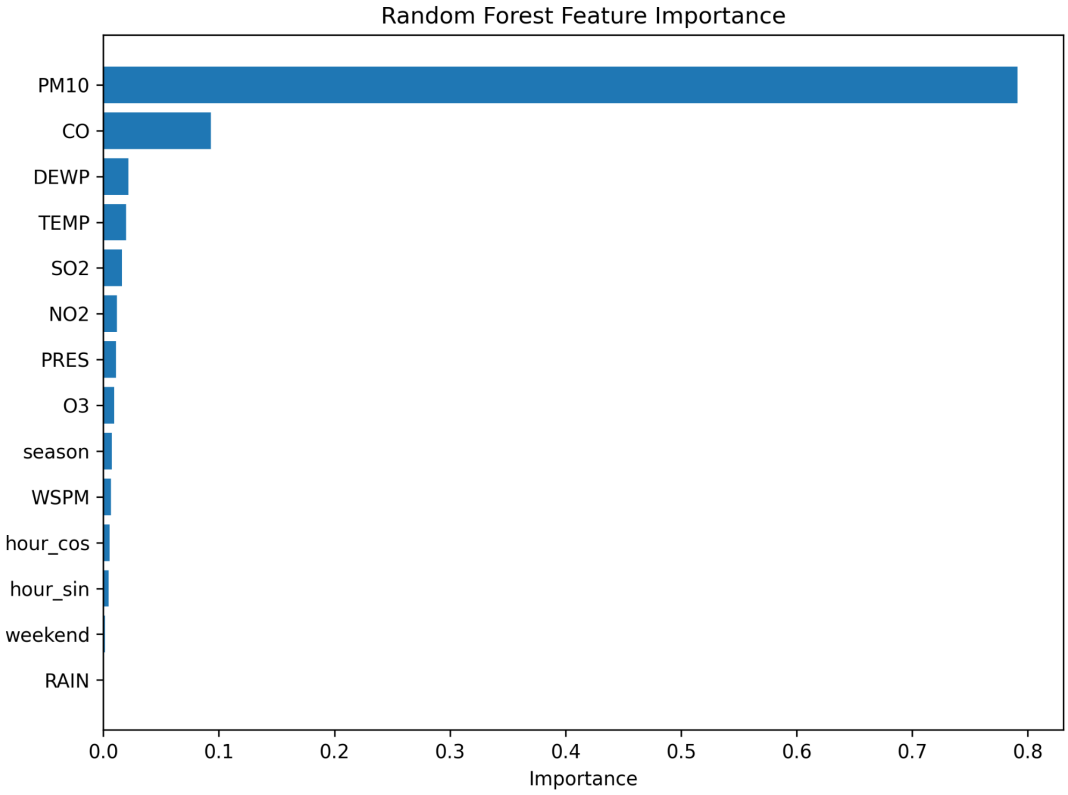


Figure 11: Feature importance scores from the tree-based model. PM_{10} was the strongest predictor, followed by CO and meteorological variables.

6 Discussion

6.1 Interpretation of Predictive Performance

The results demonstrate that ensemble tree-based methods substantially outperformed traditional linear models for $PM_{2.5}$ prediction. Extra Trees Regression achieved the strongest performance, followed closely by Random Forest Regression. This result is consistent with the nonlinear nature of atmospheric pollution, which is shaped by interactions among emissions, meteorology, atmospheric stability, transport, and chemical processes.

The weaker performance of Linear, Ridge, and Lasso Regression suggests that a simple additive linear structure is insufficient to fully describe $PM_{2.5}$ variability in this dataset. Although linear models remain valuable for interpretability, their assumptions limit their ability to capture threshold

effects and feature interactions. Ensemble models are more flexible because they partition predictor space into many decision regions and aggregate across multiple trees.

6.2 Environmental Interpretation

The strong importance of PM_{10} indicates that particulate matter fractions contain substantial shared information. CO and NO_2 are also important because they are commonly associated with combustion sources such as traffic, industrial activity, and fossil fuel use. Meteorological variables such as temperature, dew point, pressure, and wind speed influence pollutant dispersion, atmospheric mixing, and chemical transformation. Seasonal patterns and station-level differences further suggest that temporal and spatial structure are important components of urban air quality.

The seasonal analysis showed more extreme $PM_{2.5}$ concentrations during winter. This pattern may reflect increased heating-related emissions, shallower boundary layers, and more stagnant meteorological conditions. The station comparison also showed that average pollution levels differ across the monitoring network, indicating local spatial heterogeneity within the Beijing region.

6.3 Limitations

Several limitations should be considered. First, this study used observational data, so feature importance values should not be interpreted as causal effects. Second, the trained model was developed using data from Beijing and should not be assumed to generalize directly to other cities without retraining and validation. Third, the analysis predicts observed $PM_{2.5}$ concentrations rather than forecasting future air quality. A true forecasting system would require time-aware validation and lagged predictor features. Fourth, impurity-based feature importance can be influenced by correlated predictors, especially when several pollutants share emission sources.

6.4 Future Work

Future extensions could include time-series forecasting, spatial cross-validation, uncertainty quantification, SHAP or partial dependence interpretation, and comparison with additional machine learning algorithms. The workflow could also be expanded to include satellite-derived aerosol optical depth, land-use variables, traffic indicators, emissions inventories, or atmospheric reanalysis data. A more advanced version could train city-specific or station-specific models and evaluate how well a model trained in one location transfers to another.

7 Conclusion

This project developed a reproducible machine learning workflow for predicting hourly $PM_{2.5}$ concentrations using the Beijing Multi-Site Air Quality Dataset. After integrating twelve monitoring-station files and engineering temporal features, seven regression algorithms were compared using consistent evaluation metrics. Extra Trees Regression achieved the strongest predictive performance, with testing $R^2 \approx 0.946$, MAE of approximately 11.06 $\mu g/m^3$, and RMSE of approximately 18.66

$\mu g/m^3$. Five-fold cross-validation confirmed stable model performance across independent data partitions.

The analysis demonstrates that ensemble tree-based methods can effectively model nonlinear relationships between pollutant concentrations, meteorological conditions, and $PM_{2.5}$ levels. It also illustrates the importance of exploratory data analysis, careful preprocessing, model comparison, cross-validation, residual diagnostics, and transparent interpretation. While the results are predictive rather than causal, the project provides a strong portfolio example of environmental machine learning and a foundation for more advanced work in air quality forecasting, explainable artificial intelligence, and atmospheric data science.

Technical Skills Demonstrated

- **Programming:** Python, pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Plotly, Streamlit.
- **Statistics and modeling:** supervised regression, train-test splitting, ensemble learning, cross-validation, model comparison, residual diagnostics, feature importance.
- **Environmental data science:** air pollution monitoring, $PM_{2.5}$ prediction, pollutant correlations, seasonal variation, station-level spatial heterogeneity.
- **Scientific communication:** reproducible figures, dashboard deployment, structured technical reporting, LaTeX manuscript formatting.

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